Technological Progress and the Organization of Information Production in Financial Markets

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Abstract

Recent advances in AI and other general-purpose information technologies are reshaping the organization of financial information production. This paper analyzes how technological progress reshapes information production through interactions between labor markets and financial markets. Existing views emphasize that AI decentralizes information production by expanding individual capabilities. I show that this captures only one channel: technological progress can intensify competition in financial markets and eliminate the profitability of organized information production, generating a fundamentally different form of decentralization. Technological progress can concentrate or decentralize information production depending on the stage of technological development, with implications for liquidity, welfare, and technology design.

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1 Introduction

Recent advances in artificial intelligence (AI) and other general-purpose information technologies (GPTs) have dramatically increased individuals' ability to acquire and process financial information. A growing literature argues that these technologies democratize access to sophisticated information processing and reduce barriers to active trading (e.g., Chang, Dong, Margin, and Zhou, 2026), leading to the view that talented individuals and small investment firms can increasingly perform tasks that previously required large research organizations. Similarly, the recent industry reports (e.g., Bloomberg, 2026; Forbs, 2026) describe how AI-powered investment firms are replacing large analyst teams with a handful of researchers and AI systems, reconfiguring the organization of information production. Existing theories explain how technological progress affects information acquisition and market efficiency, but provide little guidance on how it reshapes the *organization* of information production.[1] This naturally raises the question: Does technological progress necessarily decentralize information production, or can it instead lead to greater concentration? More generally, why does technological progress sometimes reshape the organization of information production rather than simply improve its efficiency?

This paper develops a model of endogenous information production through financial agents' occupational choices and demonstrates that the relationship between technological progress and the organization of information production is fundamentally non-monotonic.[2] In the model, the organization determines not only who produces information, but also how much information reaches financial markets and how competition shapes informational rents. Understanding how technological progress reshapes this organization is therefore central to assessing its effects on financial markets.

The model consists of a continuum of heterogeneous agents who differ in their information-processing skills. An agent's information production capacity is jointly determined by her skill and the technology level common across all agents. Each agent first chooses between

[1]Throughout the paper, the term "organization" is used to refer to the endogenous formation of teams or firms for information production, rather than to the internal structure of firms.

[2]The main argument of the paper focuses on general-purpose information technologies, such as the Internet and generative AI, that are broadly accessible and enhance agents' ability to generate an information edge in financial markets. I abstract from special-purpose technologies, such as high-frequency trading technologies, that benefit only specific agents.

establishing a trading firm as an owner-manager and serving as a financial analyst. Firms and analysts participate in the labor market, where trading firms hire analysts to organize information production. By combining analysts' and a manager's information-production capacity, a firm generates a private signal about a risky asset. The firm then participates in the financial market, incurring a fixed participation cost, where its informational advantage depends on the manager's skill and the average skill of analysts on staff.[3] In equilibrium, occupational choice is characterized by a skill cutoff: high-skilled agents become managers and organize information production, whereas low-skilled agents optimally supply their information-processing capacity to trading firms for a uniform wage. The endogenous interaction between the analyst labor market and the financial market determines both the organization of information production and market outcomes.

Technological progress has two opposing effects on the organization of information production. First, by improving each agent's information-processing capability, it raises the return to establishing a trading firm on her own. This induces intermediate-skilled agents to establish their own trading firms instead of working as analysts, thereby reducing the supply of analysts available to incumbent firms. I refer to this force as the *capability effect*. Second, technological progress intensifies competition among informed trading firms in the financial market. Because firms fail to internalize the information revelation through their own trades (e.g., Grossman and Stiglitz, 1980), aggregate information production erodes informational rents in the financial market, weakening firms' incentives to organize information production. I refer to this force as the *competition effect*. The interaction between these two forces generates a non-monotonic evolution of the organization of information production.

At relatively low levels of technological development, information revelation is not substantial, and the capability effect dominates the competition effect. Even intermediate-skilled agents, who would normally lack the ability to organize information production and to trade on their own, can benefit from technological innovation and enter the market as managers of trading firms. Consequently, the supply of analysts declines, and each firm is forced to operate with a relatively small information-production team. Therefore information is pro-

[3]The financial market in my model combines the Kyle (1985) price-formation mechanism with competition among informed traders in the spirit of Holden and Subrahmanyam (1992), while incorporating quadratic trading costs, following Vives (2008), to prevent complete information revelation.

duced in a decentralized manner: a large number of trading firms, each supported by a small analyst team, prevail in the economy. This pattern is referred to as *capability-driven decentralization* and corresponds to the conventional view that GPTs broaden participation among a wide range of agents in the financial market by expanding individual capabilities of information production.

As technology continues to improve, however, competition among informed trading firms intensifies and erodes informational rents, thereby inducing intermediate-skilled managers to exit and return to the supply side of the labor market. Information production therefore experiences *competition-driven concentration*: only the most skilled trading firms remain active, each employing a large analyst team. Rather than broadening participation, further technological progress now reinforces concentration through the competition effect. This phase is consistent with patterns occasionally observed in financial markets, where large firms dominate talent acquisition, making it harder for smaller firms to compete (Business Insider, 2025).

Beyond a critical level of technological development, however, the competition effect becomes sufficiently strong that the hiring equilibrium ceases to exist. As informational rents are substantially eroded by competition, establishing a trading firm is no longer profitable once the fixed participation cost is taken into account.[4] Consequently, the managers' participation constraint fails, and organized information production collapses together with the analyst labor market. The economy instead transitions to a *decentralized equilibrium*, where information production and trading are carried out by stand-alone managers without organized analyst teams. Once the labor market disappears, low-skilled informed agents no longer produce information, mitigating information revelation and the competition externality. The resulting recovery in informational rents restores the profitability of stand-alone trading for sufficiently skilled agents, even though organized information production remains unsustainable. Information production therefore shifts discontinuously from analyst teams to independent traders, giving rise to a second, fundamentally different form of decentralization, referred to as *competition-driven decentralization*. Unlike capability-driven

[4]Even at zero wages, an incumbent manager would still benefit from hiring additional analysts. Rather, the remaining informational rents are too small to justify becoming a manager in the first place.

decentralization, this force arises not because individuals become more capable, but because intensified competition in the financial market eliminates the profitability of organized information production.

The model delivers three main implications. First, it identifies two fundamentally different channels through which technological progress decentralizes information production: the conventional capability effect and the competition effect arising from the financial market. Although both mechanisms generate smaller organizations and more independent traders, they have opposite implications for informational rents and market liquidity, providing a natural empirical strategy to distinguish them. Second, the model predicts that the organization of information production evolves non-monotonically with technological progress: decentralization at early stages of technological progress is followed by concentration before organized information production eventually collapses. This result provides a unified theoretical framework for understanding why technological advances can lead either to decentralized information production or to increased concentration. Finally, the model shows that organized information production need not improve agents' welfare. It induces low-skilled agents to produce information within trading firms, even though they would have remained inactive in the absence of the labor market for analysts. The additional information intensifies competition in financial markets and erodes the very informational rents that information producers seek to exploit. This finding highlights a novel trade-off between organizational information productivity and the profitability of information, suggesting an important tension in the design of information technologies, with implications for policy toward technology adoption in financial markets.

A growing literature studies how advances in AI and other information technologies affect knowledge production and information processing (Brynjolfsson, Li, and Raymond, 2023; Korinek, 2023; Chang, Dong, Margin, and Zhou, 2026). This paper contributes to this literature by studying how such technologies endogenously determine the organization of information production. In finance, Farboodi and Veldkamp (2020) show that information technology reshapes the type of information traders choose to produce. More broadly, several studies analyze information aggregation through organization (e.g., Becker and Murphy, 1992; VanZandt, 1999; Garicano, 2000; Garicano and Rossi-Hansberg, 2006). Closest to my setting, Stein (2002) shows that financial firms choose between hierarchical and decentralized information production depending on whether information can be credibly communicated up a chain of command. In these papers, the limiting force of organization is an internal cost, such as communication costs, so that technological changes affect the size of organizations in a monotonic manner. In contrast, my model embeds a competitive financial market that disciplines the expansion of organization, demonstrating non-monotonic and discontinuous evolution of organized information production.

A separate literature studies how competition among traders erodes the value of information in asset markets. Grossman and Stiglitz (1980) establish the seminal result on the interplay between information acquisition by individual traders and competition. Stein (2009) extends this logic to "crowded trades," showing the self-defeating nature of competing traders, while Glode, Green, and Lowery (2012) show that competitive over-investment in costly expertise is fully dissipated in equilibrium and can trigger a breakdown in trade when adverse selection is severe. More broadly, a large literature studies information acquisition under competitive rational-expectations equilibria (e.g., Verrecchia, 1982; Admati, 1985; Veldkamp, 2006; García and Vanden, 2009). Unlike this literature, which treats information acquisition and production as individual decisions, my model studies information production through a rival input of human capital. As a result, technological progress reshapes not only the amount of information produced, but also the organization of information production.

2 Model

Consider a three-period economy with a continuum of agents. In the first period ($t = 0$), agents with heterogeneous skills exert costly effort to acquire financial knowledge. In the second period ($t = 1$), agents sort themselves into one of two occupations: establishing a *trading firm* or working as a *financial analyst*. Each trading firm can be seen as an investment fund operated by a single owner-manager, who hires analysts to organize information production and makes trading decisions.[5] In the third period ($t = 2$), trading firms invest

[5]Throughout the paper, I use "trading firm" and "(owner-)manager" interchangeably whenever no confusion arises.

in the financial market based on information generated by analysts on staff. A single risky asset is traded in the financial market whose payoff, v, follows a normal distribution with mean $\bar{v}$ and variance σ_v^2 and is realized at the end of $t = 2$.

2.1 Skill and Knowledge

Each agent has information-production capacity, η_z, which determines the precision with which an agent can assess the magnitude of mispricing in the asset (specified below). η_z depends on the agent's idiosyncratic skill and the common information technology, as specified by the following information-production function.[6]

$$\eta_z \equiv \alpha\theta z, \tag{2.1}$$

where $z \in [0, 1]$ and $\theta \in \{0, 1\}$ represent *information-processing skill* and *financial knowledge*, respectively, and they are heterogeneous across agents. Parameter $\alpha \in [\alpha_0, 1]$ is the information productivity common across all agents, capturing the level of information technology that helps agents collect and process financial information, such as generative AI, personal computers, and the Internet.[7]

At the beginning of $t = 0$, agents independently draw information-processing skill z from a uniform distribution, $z \sim U[0, 1]$. This skill can be interpreted as the agent's inherent computational ability, which is determined exogenously and varies across agents. Upon observing z, each agent chooses whether to exert effort to acquire financial knowledge by incurring a fixed non-pecuniary effort cost $\epsilon > 0$. The acquisition of financial knowledge $\theta \in \{0, 1\}$ depends on an agent's effort choice: by exerting effort, she learns financial knowledge and obtains $\theta = 1$ with probability one, but if she does not exert effort, she obtains $\theta = 1$ only with probability $\phi < 1$. The result of knowledge acquisition is realized at the end of $t = 0$. This effort choice can be seen as studying for the CFA exam or pursuing a

[6] The main qualitative results are robust to generalizing the information production function from (2.1) to $\eta_z = \alpha\theta f(z)$, as long as $f(z)$ is increasing and weakly convex in z. I adopt the linear specification for analytical tractability. However, several analytical results may not hold under the general specification, and more importantly, the uniqueness of the equilibrium (defined below) can fail unless more specific functional forms are imposed.

[7] The common technology level is bounded from below, $\alpha_0 > 0$, to ensure the existence of an equilibrium with financial market trading.

finance degree: passing the exam or completing the program allows an agent to apply her information-processing skill z to analyze the financial market.

In what follows, z denotes an agent's *type*, and an agent who acquires financial knowledge ($\theta = 1$) is referred to as an *informed* agent, whereas an agent who fails to do so ($\theta = 0$) is referred to as an *uninformed* agent. Information about (z, θ) remains private to each agent. Also, the effort cost is positive but assumed to be vanishingly small, $\epsilon \to 0$. This assumption greatly simplifies the model without affecting the qualitative results.

2.2 Labor Market

In the second period ($t = 1$), each agent chooses her role in the financial market—either establishing a trading firm or serving as an analyst—and participates in the labor market. For simplicity, the following analyses assume that only informed agents can establish a trading firm, while Appendix A demonstrates that this restriction is without loss of generality, as it endogenously arises in equilibrium.

If a type-z agent establishes a firm, she may hire measure q_z of analysts by paying wage w and incurring a quadratic managerial cost, $\frac{\gamma}{2}q_z^2$ with $\gamma > 0$. This managerial cost serves to prevent unlimited demand for analysts. In reality, such a cost arises because hiring and deploying a larger workforce entails increasing organizational costs, making the cost increasing and convex in q_z. As specified below, each trading firm generates a private signal about the asset payoff by utilizing the aggregate information-production capacity of informed analysts together with the owner-manager's own capacity. I refer to such a hierarchical arrangement as the *organization* of information production, and the size of the organization is measured by q_z. Note that a trading firm can always choose not to hire analysts ($q_z = 0$) and instead engage in independent information production by using only the manager's information-processing capacity.

Alternatively, an agent may choose to be an analyst. Such an agent receives wage income w and is hired by a trading firm through a competitive labor market.[8] If the analyst is informed, then she contributes to the firm's information production, because she has $\eta_z > 0$;

[8] An analyst may trade the asset through independent information production, but such an agent is classified as a trading firm with a stand-alone manager according to the definition.

if she is uninformed, $\eta_z = 0$, and she makes no contribution.[9] In what follows, I denote the set of agents who establish trading firms as owner-managers by M, and the set of agents who work as financial analysts by A.

The labor market for financial analysts is competitive and anonymous. The competitiveness means that the wage w is determined by the market clearing condition, and the anonymity implies that firms hire analysts without knowing which analysts are informed. Importantly, analysts on the supply side of the labor market receive the pooling wage w regardless of their financial knowledge, so that w can be seen as the upfront signing bonus paid before analysts perform the actual information-production tasks.

2.3 Information

Suppose that a trading firm of type z has hired measure q_z of analysts. It utilizes the aggregate information-production capacity, including its own, to generate a private signal about the asset payoff. In particular, I define the firm's aggregate information-production capacity by

$$\tau_z \equiv \eta_z + q_z \mu_A, \tag{2.2}$$

where

$$\mu_A \equiv \mathrm{E}[\eta_z | z \in A] \tag{2.3}$$

represents the average information-production capacity of analysts hired in the labor market. The first term of τ_z captures the firm manager's own contribution to the total capacity, and the second term represents the total contribution of hired analysts. By leveraging τ_z, the firm observes a private signal about the asset payoff, $s_z = v + \xi_z$, where the informativeness of this signal is directly proportional to τ_z. In particular, the noise term ξ_z follows a normal distribution with mean zero and variance $\mathrm{Var}[\xi_z] \equiv \sigma_v^2 \frac{1-\tau_z}{\tau_z}$, implying that the signal informativeness becomes

$$\frac{\mathrm{Var}[v]}{\mathrm{Var}[s_z]} = \tau_z. \tag{2.4}$$

[9] If we instead assume uninformed analysts' knowledge level is $\theta = \theta_0$ with $0 < \theta_0 < 1$, they can make a positive contribution to information production. This alternative assumption does not change the main results of the paper, while complicating the analysis.

A larger aggregate information-production capacity τ_z reduces the noise variance of the signal s_z and yields more precise private information to the firm. The firm observes the signal s_z at the beginning of the third period ($t = 2$) and makes its investment decision based on s_z.

2.4 Financial Market

The financial market in the last period ($t = 2$) has three types of participants: trading firms, a noise trader, and a competitive market maker.[10] I assume that each trading firm incurs a fixed participation cost $\kappa > 0$ to trade in the financial market, capturing fixed expenditures associated with establishing research infrastructure, meeting compliance requirements, and maintaining operational setup (e.g., Vissing-Jørgensen, 2002).

Upon participating in the financial market, a trading firm of type z places a market order for $x_z \in \mathbb{R}$ units of the asset, where the firm faces a quadratic transaction cost $\frac{\rho}{2}x_z^2$ with $\rho > 0$. This cost can be seen as capturing risk aversion or inventory pressure and ensures that the optimal investment is finite (Vives, 2008). The noise trader places an exogenous and random market order for u units of the asset, where u follows a normal distribution with mean zero and variance σ_u^2. The market maker observes the total order flow, $y \equiv \int_{z\in M} x_z dz + u$, and sets the execution price of the asset, p. Due to competition, the price is determined by the semi-strong efficient level:

$$p = \mathrm{E}[v|y]. \tag{2.5}$$

After all orders are executed, the asset payoff v is realized, and each firm earns trading profit $\pi_z \equiv (v - p)x_z - \frac{\rho}{2}x_z^2$.

[10] Following the convention in the market microstructure literature, I assume that there are competitively many market makers off the equilibrium path, and only one of them executes incoming orders on the equilibrium path. This setting ensures the competitive asset price because, if the active market maker deviates from the competitive price, then other market makers would quote a better price.

2.5 Utility and Equilibrium

A type-z agent obtains the following utility by establishing a trading firm, hiring q_z analysts, and trading x_z units of the asset:

$$U_{z,M} = \pi_z - wq_z - \frac{\gamma}{2}q_z^2 - \kappa,$$

where the second and the third terms are the wage payment and the managerial cost, and the last term captures the fixed participation cost of the financial market. If an agent becomes an analyst, in contrast, her utility stems from the wage payment,

$$U_{z,A} = w.$$

The utility of the noise trader directly reflects the price impact of her order u and is formalized later in Section 5. The market maker breaks even by trading at the competitive price.

The equilibrium of the model is defined as follows:

Definition 1. The equilibrium of the model consists of $(A, M, \{x_z, q_z\}_{z\in M},\ p,\ w)$ and individual agents' effort choices, such that; (i) each agent chooses her effort to maximize her expected utility, $\mathrm{E}[U_{z,A}]$ or $\mathrm{E}[U_{z,M}]$, incorporating the effort cost ϵ; (ii) the sets of analysts and trading firms, A and M, are determined by each agent's occupational choice that maximizes her expected utility, $\mathrm{E}[U_{z,A}]$ or $\mathrm{E}[U_{z,M}]$; (iii) a trading firm of type z chooses the trading quantity x_z to maximize the conditional expected trading profit $\mathrm{E}[\pi_z|s_z]$; (iv) it chooses the demand for analysts q_z to maximize the expected utility $\mathrm{E}[U_{z,M}]$; (v) the asset price p is determined by the efficiency condition in (2.5); and (vi) the wage w is determined by the market-clearing condition of the labor market.

Finally, I impose the following parameter conditions:

Assumption 2. $\sigma_v > \max\left\{\sqrt{\frac{\rho\kappa}{\alpha_0}}, \frac{4\rho\sigma_u}{\phi}\right\}$.

This assumption guarantees the existence of at least one type of equilibrium (specified below) with active trading firms by ensuring that the value of acquiring and trading on private information is sufficiently high.

3 Equilibrium

3.1 Financial Market

The model is solved by backward induction. I guess and later verify that a type-z trading firm in the financial market ($t = 2$) takes the following linear trading strategy.

$$x_z = \beta\tau_z(s_z - \bar{v}), \tag{3.1}$$

where β is determined in equilibrium and is referred to as the trading intensity. In equation (3.1), $\tau_z(s_z - \bar{v})$ represents the *ex-ante* mispricing in the asset: the market maker's prior expectation is $\mathrm{E}[v] = \bar{v}$, and she would quote $\bar{v}$ without observing the order flow, while the informed trading firm's conditional expectation is $\mathrm{E}[v|s_z] = \bar{v} + \tau_z(s_z - \bar{v})$. Thus, the difference between them, $\tau_z(s_z - \bar{v})$, can be seen as the asset's mispricing, and its magnitude is scaled by the firm's aggregate information-production capacity, τ_z. The strategy in (3.1) implies that the firm exploits this mispricing with intensity β.

According to the trading strategy in (3.1), the aggregate order flow from trading firms is defined as

$$X \equiv \int_{z\in M} x_z dz = \beta\tau(v - \bar{v}), \tag{3.2}$$

where

$$\tau \equiv \int_{z\in M} \tau_z dz. \tag{3.3}$$

As τ captures the *economy-wide* information-production capacity that all active agents devote to analyze the risky asset, it is referred to as the *aggregate information quality* in the following discussion. From the market maker's perspective, the aggregate order flow, $y = X + u$, is linear in v.[11] The standard filtering leads to the following efficient price.[12]

$$p = \bar{v} + \lambda y, \tag{3.4}$$

[11]The condition for the exact law of large numbers is satisfied in this model, as $\int_{z\in M} \mathrm{Var}[\xi_z]dz < \infty$, and idiosyncratic noise terms $\{\xi_z\}_{z\in M}$ in equation (3.2) average out in the cross section, implying $\int_{z\in M} \xi_z dz = 0$ almost surely.

[12]It holds that $p = \mathrm{E}[v|y] = \bar{v} + \frac{\mathrm{Cov}[v,y]}{\mathrm{Var}[y]}(y - \mathrm{E}[y])$, where $\lambda = \frac{\mathrm{Cov}[v,y]}{\mathrm{Var}[y]}$ yields the result in (3.5).

where

$$\lambda = \frac{\beta\tau\sigma_v^2}{\beta^2\tau^2\sigma_v^2 + \sigma_u^2}. \tag{3.5}$$

The coefficient λ represents the price impact of order flow and, following Kyle (1985), λ is used as the measure of illiquidity: a high λ implies an illiquid financial market.

Since the hiring and the participation costs are sunk at the trading stage, a trading firm of type z chooses its trading strategy x_z to maximize the expected net trading profit:

$$\max_{x_z} \mathrm{E}[(v-p)x_z|s_z] - \frac{\rho}{2}x_z^2, \tag{3.6}$$

where the firm incorporates the pricing rule in (3.4) and the aggregate order flow (3.2), while its price impact is zero since each firm's measure is infinitesimal. The first-order condition of (3.6) yields the following optimal strategy.

$$x_z = \frac{1-\lambda\beta\tau}{\rho}\tau_z(s_z - \bar{v}),$$

suggesting that the guess in (3.1) is correct with the trading intensity

$$\beta = \frac{1-\lambda\beta\tau}{\rho}. \tag{3.7}$$

Solving equations (3.5) and (3.7) yields β and λ as functions of τ:

Lemma 3. *(i) The optimal trading strategy of trading firms, β, is determined by the unique solution to the following fixed-point problem and is monotonically decreasing in the aggregate information quality τ.*

$$\rho\beta = \frac{\sigma_u^2}{\beta^2\tau^2\sigma_v^2 + \sigma_u^2}. \tag{3.8}$$

(ii) By applying the solution β, the price impact, λ, is determined by equation (3.5) and exhibits a single-peaked reaction to τ.

The price impact exhibits a hump-shaped curve against τ. Intuitively, τ captures the total information advantage of active trading firms, and an increase in τ imposes a larger adverse selection cost on the market maker, thereby inducing her to set a larger price impact. However, as τ becomes sufficiently large, the order flow reveals excessive information about v

from the perspective of individual trading firms, because each firm ignores the information-revelation externality generated by its own trades. Anticipating this information revelation, the market maker places less weight on the order flow when inferring the asset value, leading to a lower price impact.[13] To avoid moving the price too much, each manager trades less aggressively by lowering β when τ becomes larger.[14]

Applying (3.5) and (3.7), the expected trading profit of the type-z trading firm, conditional on s_z, is computed as

$$\pi_z = \rho\beta^2\tau_z^2(s_z - \bar{v})^2.$$

Intuitively, the firm's informational advantage is $\tau_z(s_z - \bar{v})$, and each unit traded is expected to earn this profit margin. Since the firm scales the trading strategy x_z by its informational advantage as well, the total expected profit becomes quadratic in $\tau_z(s_z - \bar{v})$. Also, by taking the unconditional expectation of π_z,

$$\bar{\pi}_z = \mathrm{E}[\pi_z] = \rho\sigma_v^2\beta^2\tau_z. \tag{3.9}$$

Therefore, the firm's *ex-ante* expected trading profit (prior to observing s_z) becomes proportional to the aggregate information-production capacity τ_z of the firm's organization.

3.2 Labor Market

This subsection explores how the organization of information production is shaped through agents' occupational choices in the labor market.

[13] While λ exhibits a single-peaked reaction to τ, the price impounds information v with coefficient $\beta\lambda\tau$. This represents the price sensitivity to v and is monotonically increasing in τ (see [3.5]).

[14] In the Kyle (1985) model, the insider trades as a monopolist in the financial market by fully incorporating the information revelation via her trading activity. In this model, in contrast, informed managers do not internalize this effect, because individual managers are infinitesimal and have no price impact. As a result, they reveal too much information, causing λ to negatively react to τ.

3.2.1 Trading Firm's Demand for Analysts

Given the expected trading profit $\bar{\pi}_z$ in (3.9), a type-z trading firm chooses its demand for analysts by solving the following problem.[15]

$$\max_{q_z \geq 0} \bar{\pi}_z - wq_z - \frac{\gamma}{2} q_z^2.$$

Since the firm is infinitesimal, it does not internalize the impact of its analyst demand q_z on the aggregate information quality τ. Noting that the firm's information-production capacity is $\tau_z = \alpha z + q_z \mu_A$, the first-order condition yields the following optimal analyst demand:

$$q^* = \max \left\{ \frac{\rho \sigma_v^2 \beta^2 \mu_A - w}{\gamma}, 0 \right\}. \tag{3.10}$$

The marginal information productivity of analysts, μ_A, is independent of the firm's type z, making the optimal demand q^* common across all firms. Equation (3.10) suggests the possibility of two types of equilibria: one with active hiring, $q^* > 0$, and the other with no labor market activity, $q^* = 0$. In what follows, I focus on the active hiring equilibrium and leave the analysis of the non-hiring equilibrium for Subsection 3.3.

In the hiring equilibrium, the aggregate demand for analysts is given by

$$Q = \int_{z \in M} \left(\frac{\rho \sigma_v^2 \beta^2 \mu_A - w}{\gamma} \right) dz. \tag{3.11}$$

The type-z firm expects to obtain the following indirect utility by hiring q^* analysts and paying the fixed participation cost:

$$\mathrm{E}[U_{z,M}] = \alpha \rho \sigma_v^2 \beta^2 z + \frac{(\rho \sigma_v^2 \beta^2 \mu_A - w)^2}{2\gamma} - \kappa. \tag{3.12}$$

The first term represents the expected net trading profit arising from the firm's own information-processing skill z, and the second term is that arising from hiring analysts, net of the wage

[15] Note that the market-participation constraint is redundant. Any agent who chooses to become a manager must earn a higher payoff net of the participation cost than the analyst wage w. Since $w > 0$, it follows that the manager's payoff before paying the participation cost must exceed the cost itself, implying that market participation is always optimal.

payment and the managerial cost.

3.2.2 Occupational Choice and Effort

A type-z informed agent compares the expected utility of a trading firm $\mathrm{E}[U_{z,M}]$ in (3.12) with the analysts' wage income w: she establishes a trading firm if and only if $\mathrm{E}[U_{z,M}] \geq w$.[16] Since $\mathrm{E}[U_{z,M}]$ in equation (3.12) increases with an agent skill z, this comparison is characterized by a cutoff, z^*. In particular, informed agents with relatively high skills, $z \geq z^*$, choose to establish firms, while both uninformed agents with $z \geq z^*$ and all agents with relatively low skills, $z < z^*$, become financial analysts.

Moreover, the cutoff z^* characterizes the effort choice as well. Namely, agents with $z < z^*$ would not found a trading firm, even if they acquire financial knowledge, and end up receiving the pooling wage w regardless of their effort choice. Since the effort choice does not affect their expected utility, while exerting effort is costly, they optimally choose not to exert effort. In contrast, the occupational choice of agents with relatively high skill, $z \geq z^*$, depends on the result of their knowledge acquisition: if an agent becomes informed, she chooses to establish a trading firm, while she becomes an analyst otherwise. As $\mathrm{E}[U_{z,M}] \geq w$ holds for these agents, the marginal benefit of effort is positive. As the effort cost is diminishing ($\epsilon \to 0$), they optimally choose to exert effort.

Lemma 4. *There is a unique cutoff z^* that solves the following indifference condition:*

$$\alpha\rho\sigma_v^2\beta^2 z^* + \frac{(\rho\sigma_v^2\beta^2\mu_A - w)^2}{2\gamma} = w + \kappa. \tag{3.13}$$

Agents with type $z \geq z^*$ exert effort and establish trading firms, while those with type $z < z^*$ do not exert effort and become analysts.

Lemma 4 implies that the set of analysts is characterized by $A = \{z|z < z^*\}$, while the set of trading firms is $M = \{z|z \geq z^*\}$. Hence, the organization of information production in the model is characterized by the cutoff z^*: agents with relatively low skills ($z \in A$) are hired by trading firms operated by managers with relatively high skills ($z \in M$). Analysts

[16] An agent is assumed to establish a firm when she is indifferent between the two occupational roles, i.e., $\bar{U}_{z,M} = w$. This tie-breaking rule is, however, innocuous to the results.

produce information about the risky asset under the supervision of these trading firms. As the cutoff z^* increases, a larger mass of agents becomes analysts, and information production is concentrated in a smaller set of high-skilled trading firms. Conversely, a decline in z^* induces more agents to establish a firm by exerting effort. The information production is more dispersed across a large number of firms with a wide range of skills, where each firm organizes a small analyst team.

3.2.3 Labor Market Clearing

According to Lemma 4, the aggregate analyst supply is given by $S \equiv \int_0^{z^*} dz = z^*$, and equation (3.11) leads to the following labor market clearing condition:

$$\frac{\rho\sigma_v^2\beta^2\mu_A - w}{\gamma}(1 - z^*) = z^*. \tag{3.14}$$

In equilibrium, the wage and the cutoff jointly adjust so that each firm's demand for analysts becomes $q^* = \frac{z^*}{1-z^*}$. Since the indifference condition (3.13) always has a unique solution $z^* > 0$, the labor market clearing condition (3.14) ensures $q^* > 0$ in the hiring equilibrium. However, condition (3.14) also pins down the wage as

$$w(z^*) = \rho\sigma_v^2\beta^2\mu_A - \gamma\frac{z^*}{1 - z^*}, \tag{3.15}$$

leading to the possibility that z^* induces $w(z^*) \leq 0$. The occupational and effort choices analyzed so far are based on a positive wage, and $w(z^*) \leq 0$ cannot motivate analysts to participate in the labor market.[17] Such an equilibrium with the labor market breakdown is analyzed in Subsection 3.3, and the following argument focuses on the case with $w(z^*) > 0$.

3.2.4 Analyst Quality and Information Production

On the supply side, informed and uninformed analysts are pooled together with mass ϕ and $1 - \phi$, and the anonymity of the labor market leads to the quality uncertainty from the

[17] To avoid indifference between active information production and inactivity when $w = 0$, we can impose an arbitrarily small common disutility of activity, which vanishes in the limit. Since it is common across managers and analysts, it does not affect occupational sorting in the hiring equilibrium.

demand-side perspective. However, the average information productivity of analysts, μ_A in equation (2.3), becomes deterministic due to the law of large numbers:

$$\mu_A = \mathrm{E}\left[\alpha z\theta | z < z^*\right] = \frac{\alpha\phi z^*}{2}. \tag{3.16}$$

As the cutoff z^* increases, agents with relatively high skills choose to be analysts, thereby improving the average quality of the analyst pool in the labor market.

Also, the aggregate information quality in the financial market (3.3) becomes

$$\tau = \int_{z\in M} (\alpha z + q^*\mu_A)dz = \alpha\frac{1-(1-\phi)z^{*2}}{2}, \tag{3.17}$$

where $\int_{z\in M} q^* dz = z^*$ holds due to the labor market clearing condition. Interestingly, the aggregate information quality τ is decreasing in z^*, showing the opposite behavior compared to the analysts' information productivity μ_A.[18] When z^* increases, a larger fraction of agents remains in the "no effort" regime, where these agents' expected productivity is only ϕz rather than z. The information production loss from these additional agents dominates the compositional effect in the labor market that raises the conditional average μ_A, causing a decline in τ. In what follows, I denote the total information quality as $\tau(z^*)$ and the trading intensity $\beta(z^*) = \beta(\tau(z^*))$ to specify their dependence on z^*, where Lemma 3 and equation (3.17) imply that $\beta(z^*)$ is monotonically increasing in z^*.

By incorporating (3.14)–(3.17), the hiring equilibrium is characterized by the following:

Proposition 5 (Hiring Equilibrium)**.** *There is at most one hiring equilibrium characterized by the cutoff z^*. The cutoff is the unique solution to the following condition*

$$\alpha\rho\sigma_v^2\beta^2(z^*)z^*\left(1-\frac{\phi}{2}\right) + \gamma\frac{z^*\left(1-\frac{z^*}{2}\right)}{(1-z^*)^2} = \kappa, \tag{3.18}$$

where $\beta(z^*)$ is the unique solution to (3.8).

The LHS of equation (3.18) represents the expected net trading profit of a type-z^* trading firm, after incorporating the hiring cost and the opportunity cost of serving as an analyst

[18]See Fukui (2018) for the similar argument in the context of capital markets with adverse selection.

(i.e., the wage income), while κ on the RHS is the participation cost of the financial market. As the LHS is monotonically increasing in z^*, condition (3.18) ensures a unique solution $z^* \in (0, 1)$.[19]

3.3 Decentralized Equilibrium

The solution z^* to (3.18) constitutes the hiring equilibrium only if the wage in (3.15) is strictly positive. As elaborated in Section 4, such an equilibrium may not exist. Even in the absence of the hiring equilibrium, however, the other equilibrium with no labor market activity always exists:

Proposition 6 (Decentralized Equilibrium)**.** *There always exists an equilibrium with no labor market activity characterized by the cutoff z_d^* as the unique solution to the following indifference condition:*

$$\alpha\rho\sigma_v^2\beta_d^2(z_d^*)z_d^* = \kappa, \tag{3.19}$$

where $\beta_d(z_d^*)$ is the unique solution to

$$\rho\beta = \frac{\sigma_u^2}{\beta^2\tau_d^2(z_d^*)\sigma_v^2 + \sigma_u^2}$$

with

$$\tau_d(z_d^*) = \frac{\alpha(1 - z_d^{*2})}{2}. \tag{3.20}$$

Agents with $z \geq z_d^*$ exert effort and establish trading firms, while those with $z < z_d^*$ do not exert effort and stay inactive in the subsequent periods.

As in the hiring equilibrium, agents with relatively high skills ($z \geq z_d^*$) exert effort and become informed, while those with low skills ($z < z_d^*$) do not. Low-skilled agents stay inactive in the subsequent periods regardless of the result of knowledge acquisition because of the absence of the labor market. Moreover, high-skilled informed agents do not hire an analyst in this equilibrium; instead they join the financial market as stand-alone traders. The equilibrium in Proposition 6 is referred to as the *decentralized* equilibrium, because no

[19]The LHS of (3.18) becomes zero at $z^* = 0$ and explodes to positive infinity as $z^* \to 1$. Hence, condition (3.18) always has a unique solution in $z^* \in (0, 1)$.

organized structure for information production emerges, and all information is produced and utilized in trading through stand-alone managers operating trading firms.

The aggregate information quality, τ_d, has a slightly different structure from that in the hiring equilibrium (τ in equation [3.17]). Due to the absence of the labor market, agents who do not exert effort but obtain financial knowledge stay inactive in the decentralized equilibrium, whereas they would have produced information as analysts if the hiring equilibrium had existed and the labor market had been active. This difference highlights the information loss in the decentralized equilibrium, captured by $\tau_d(z) < \tau(z)$ at the same level of z.

As the decentralized equilibrium always exists, the existence of a hiring equilibrium necessarily implies equilibrium multiplicity, a feature common to models of adverse selection in asset markets (e.g., Kurlat, 2013; Fukui, 2018). In such a situation, the following result holds.

Lemma 7. *The decentralized equilibrium is unstable and not robust to small changes in parameters when it coexists with the hiring equilibrium.*

The instability of the decentralized equilibrium under equilibrium multiplicity is analogous to the argument in Wilson (1980), Stiglitz and Weiss (1981), and Kurlat (2013). On one hand, hiring a small mass of analysts is beneficial for a trading firm when $w \to 0$, suggesting that they are willing to hire at very low wage levels. On the other hand, informed low-skilled agents ($z < z^*$) are willing to serve as analysts whenever $w > 0$, as they earn zero utility if they stay inactive. Thus, such deviation (i.e., an active labor market) is supported as a hiring equilibrium whenever z^* in (3.13) guarantees $w(z^*) > 0$, meaning that the decentralized equilibrium is unstable. When z^* induces $w(z^*) \leq 0$, in contrast, the decentralized equilibrium is a unique and stable outcome. Because the multiplicity of equilibria is not the focus of this study, I follow the above literature and assume that the economy coordinates on the hiring equilibrium whenever it coexists with the decentralized one.

4 Technology and Organized Information Production

This section explores the impact of technology improvement (increases in α) on the organization of information production. I first focus on the behavior of the hiring equilibrium in Subsection 4.1. In Subsection 4.2, its existence condition is specified, and the phase transition between the hiring and the decentralized equilibrium is characterized.

4.1 Hiring Equilibrium

4.1.1 Information Externality

The indifference condition in equation (3.13) suggests that the information-production technology α influences the cutoff z^* by changing the expected *profit margin* of information production, which is defined as

$$R(z^*, \alpha) \equiv \alpha \rho \sigma_v^2 \beta^2(\tau(z^*, \alpha)), \tag{4.1}$$

where I denote $\beta(z^*) = \beta(\tau(z^*, \alpha))$ to underline the dependence of the trading strategy on the technology level α. Namely, a type-z trading firm leverages its own skill to earn $R(z^*, \alpha)$. On top of that, each unit of information-production capacity acquired through hiring analysts also earns $\rho\sigma_v^2\beta^2\mu_A = \frac{\phi}{2}R(z^*, \alpha)$. The profit margin is common across all firms, as the price impact in the financial market is determined by the total information quality τ.

Lemma 8. *With the cutoff z^* being fixed, a trading firm's expected profit margin R takes a single-peaked curve with respect to α. There is a unique tipping point α^*, such that, $\frac{\partial R(\alpha, z^*)}{\partial \alpha} \gtreqless 0 \Leftrightarrow \alpha^* \gtreqless \alpha$, where α^* is defined by the unique solution to (B.4) in Appendix B.*

Firstly, α has *the capability effect*: it directly increases the profit margin R by expanding the information-production capability of a firm's organization, as captured by the first coefficient in (4.1). The expansion of information production, however, has an indirect negative effect on R, which is referred to as *the competition effect.* Namely, an increase in α leads to a higher aggregate information quality τ, and the order flow $y = X + u$ reveals substantial information about v to the market maker. Consequently, each firm must hold back from trading aggressively by lowering β and settle for a smaller profit margin. In a monopolistic informed-trader model such as Kyle (1985), the trader internalizes the effect of its trading intensity on price impact and chooses β accordingly. In the Kyle (1985) equilibrium, this choice offsets the non-monotonic response of λ to β, whereas in our competitive financial

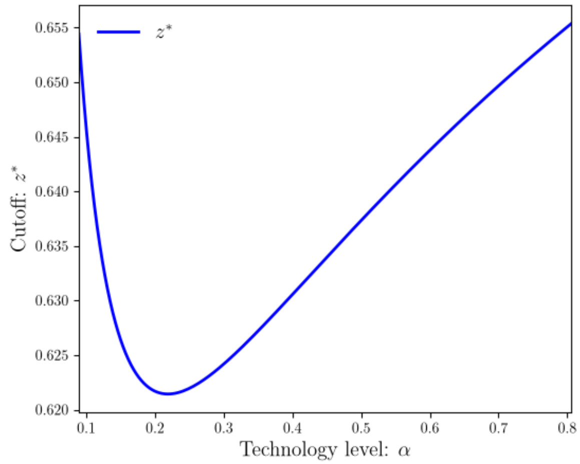

Figure 1: The hiring-equilibrium cutoff z^*

Note: This figure plots the cutoff in the hiring equilibrium using the following parameter values: $\phi = 0.2, \kappa = 2.5, \gamma = 0.12, \rho = 0.1, \sigma_v^2 = 7.0$, and $\sigma_u^2 = 1.0$.

market, individual firms do not internalize this effect (e.g., Grossman and Stiglitz, 1980; Vives, 2008). Consequently, the indirect effect of α on the profit margin can either reinforce or offset its direct capability effect, leading to an ambiguous reaction of R to changes in α.

4.1.2 Information Production

Reflecting Lemma 8, the cutoff z^* exhibits the following reaction to changes in α.

Proposition 9. *The cutoff z^* exhibits a U-shaped reaction to increases in α. With the tipping point α^* defined in Lemma 8, it holds that $\frac{dz^*}{d\alpha} \gtrless 0 \Leftrightarrow \alpha \gtrless \alpha^*$.*

Figure 1 visualizes the result in Proposition 9. If the profit margin R increases with α ($\alpha < \alpha^*$), all firms experience an exogenous upward shift in their expected utility, and agents with skill levels slightly lower than z^* are now willing to establish a trading firm, thereby lowering the cutoff in condition (3.13). If, in contrast, R declines with α ($\alpha > \alpha^*$), then agents with skill levels slightly higher than the cutoff cease to operate trading firms and stop exerting effort, so that the cutoff z^* increases. As a consequence, the cutoff z^* exhibits a U-shaped curve against α, as opposed to the profit margin R.

This result implies a non-monotonic evolution of the organization of information production. When the technology level is relatively low ($\alpha < \alpha^*$), technological progress causes

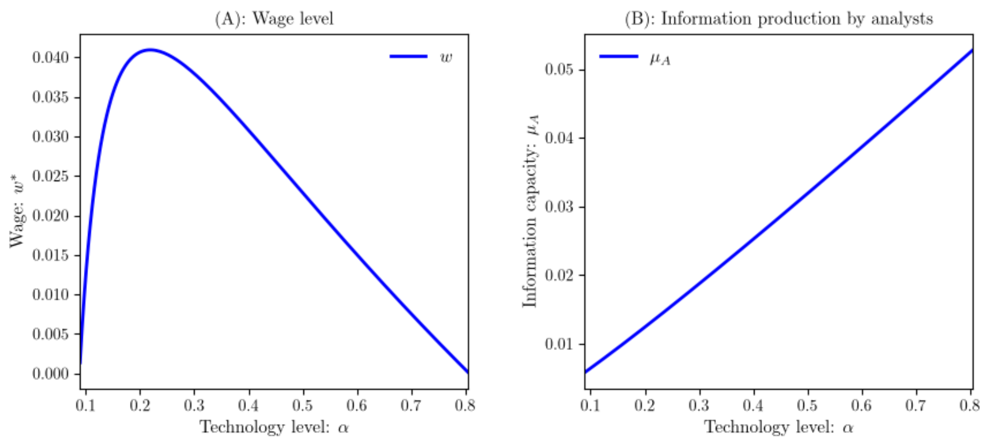

Figure 2: The labor market

Note: This figure plots the wage level, w, and the average information production by analysts, $\mu_A = \frac{\alpha\phi z^*}{2}$, in the hiring equilibrium. The parameter setting is the same as that used in Figure 1.

decentralization of information production: more agents, including those with intermediate and relatively low skills, become managers and seek to establish trading firms. As each firm can hire a smaller mass of analysts ($q^* = \frac{z^*}{1-z^*}$ holds in the hiring equilibrium), the economy consists of a large number of small-sized information production organizations. This evolution is triggered by the expansion of individual information-production capability and is referred to as *capability-driven decentralization*.

In contrast, when technology becomes mature ($\alpha > \alpha^*$), a further increase in the technology level induces *competition-driven concentration* of organized information production. Namely, an increase in α lowers the profit margin R and raises the cutoff z^*, pushing a marginal trading firm to stop operating and to switch to the analyst job. This phase exhibits a reversal toward a concentrated structure, where a small number of high-skilled trading firms organize information production on a large scale.

4.1.3 Labor Market

The labor market for financial analysts exhibits the following reaction to technological progress:

Proposition 10. *(i) The analysts' wage, w, takes a single-peaked curve with respect to the technology level α, that is, $\frac{dw}{d\alpha} \gtreqless 0 \Leftrightarrow \alpha^* \gtreqless \alpha$, where the tipping point α^* is the same as that* *in Proposition 9.*

(ii) The average information-production capacity of analysts, μ_A in equation (3.16), monotonically increases with α.

Figure 2 illustrates the results in Proposition 10. The non-monotonic reaction of the wage is a direct consequence of the U-shaped cutoff z^*. For $\alpha < \alpha^*$, as α increases, the cutoff z^* declines, and a larger mass of trading firms participate in the demand side of the labor market, while the analyst supply shrinks. Since the labor market must clear, the intensive margin of the analyst demand must also shrink, as captured by $q^* = \frac{z^*}{1-z^*}$. To support this reaction, the corresponding wage level increases. As the opposite argument holds for $\alpha > \alpha^*$, the wage level takes a single-peaked curve against α.

Along with this change in the composition of the labor market, the average information productivity of analysts on the supply side monotonically increases, as Panel B in Figure 2 illustrates. Firstly, a higher technology level α directly expands the information-production capacity of each analyst, thereby increasing μ_A. Secondly, the ambiguous reaction of z^* may or may not improve μ_A. However, this second channel is an indirect effect of α through the labor market and cannot dominate its first direct effect, resulting in the monotonically increasing information productivity of analysts.

4.1.4 Financial Market

The financial market exhibits the following reaction to the technological progress:

Proposition 11. *(i) The aggregate information quality τ, defined by equation (3.17), is monotonically increasing in the technology level α.*

(ii) Firms' trading intensity β is monotonically decreasing in α.

(iii) The price impact λ takes a single-peaked curve against α, that is, $\frac{d\lambda}{d\alpha} \gtrless 0 \Leftrightarrow \alpha^* \gtrless \alpha$, where the tipping point α^* is the same as that in Proposition 9.

Although the cutoff z^* and the organization of information production exhibit non-monotonic reactions to technological progress, the total information production, measured by τ, monotonically increases with α due to the same mechanism that drives μ_A. Responding to this heightened information production τ, the trading intensity β monotonically declines,

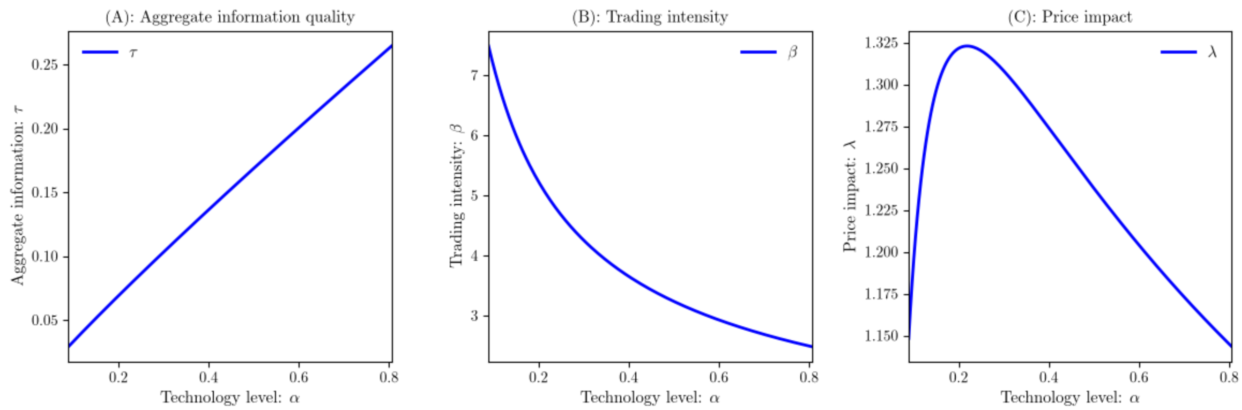

Figure 3: The financial market

Note: This figure plots the aggregate information quality, τ, the trading intensity, β, and the price impact, λ, in the hiring equilibrium. The parameter setting is the same as that used in Figure 1.

as each trading firm seeks to avoid revealing too much information about the asset fundamentals. As this reaction of trading firms cannot internalize the information externality mentioned in Subsection 4.1.1, the price impact exhibits a single-peaked curve against α. The implications of these financial-market responses are discussed further in Subsection 4.3, following the characterization of the phase transition.

4.2 Collapse of Organized Information Production

As mentioned in Subsection 3.3, the hiring equilibrium may disappear when the solution z^* of the indifference condition (3.13) cannot support a positive wage, $w > 0$. The following analyses explore what technology levels support the hiring equilibrium and, when it collapses, how the economy transitions from the hiring to the decentralized equilibrium. To focus on interesting scenarios where the hiring equilibrium exists under a certain range of α, the following analyses assume that condition (C.2) in Appendix C holds.[20]

4.2.1 Technology Thresholds

Due to the hump-shaped reaction of the wage w to the technology level α (Proposition 10), w can be negative when α is either sufficiently small or sufficiently large, leading to the

[20]If this condition is violated, the decentralized equilibrium becomes a unique outcome for all parameter values, and the model is unable to analyze organized information production.

following existence condition of the hiring equilibrium.

Proposition 12. *(i) There are two thresholds of the technology level,* α_l *and* α_h $(> \alpha_l)$*, defined by the following equation: for* $j \in \{l, h\}$*,*

$$\alpha_j \beta^2(\tau(z_0, \alpha_j)) = \frac{2\gamma}{\phi\rho\sigma_v^2(1 - z_0)}, \tag{4.2}$$

with $z_0 \equiv \frac{\phi\kappa+\gamma-\sqrt{\gamma\kappa\phi^2/2+\gamma^2}}{\phi\kappa+2\gamma(1-\phi/4)}$*. The hiring equilibrium exists if, and only if,* $\alpha \in (\alpha_l, \alpha_h)$*. If* $\alpha \notin (\alpha_l, \alpha_h)$*, then the decentralized equilibrium is a unique outcome.*

(ii) At the phase transition from the hiring equilibrium to the decentralized equilibrium, the cutoff experiences a discontinuous jump from z^* *to* z_d^**. It involves a downward jump* $(z^* > z_d^*)$ *if, and only if,*

$$\frac{\beta(\tau_d(z_0, \alpha_j))}{\beta(\tau(z_0, \alpha_j))} > \frac{\kappa\phi}{2\gamma} \frac{\gamma\left(1 - \frac{\phi}{2}\right) + \sqrt{\gamma\kappa\phi^2/2 + \gamma^2}}{\phi\kappa + 2\gamma\left(1 - \frac{\phi}{4}\right)}. \tag{4.3}$$

Otherwise, the transition from the hiring to the decentralized equilibrium involves an upward jump in the cutoff.

Figure 4 plots the equilibrium variables against the entire range of α. The figure focuses on the parameter values that satisfy condition (4.3), generating $z^* > z_d^*$, while Figure 6 in Appendix C illustrates the phase transition with $z_d^* > z^*$.

Proposition 12 attests that the hiring equilibrium exists only when the technology level α is intermediate. Firstly, the collapse of the hiring equilibrium under low levels of α is a direct consequence of insufficient information production. In such a situation, agents' information-production capacity tends to be small (given their skills, z), and the profit margin in the financial market becomes too small to sustain organized information production.

Secondly, even high levels of α can eliminate the hiring equilibrium. This is due to the competition effect: excessive information production intensifies competition among firms and erodes profit margins despite the superior technology (Lemma 8). Consequently, when the technology level is sufficiently high, the model exhibits *competition-driven decentralization* of information production. That is, even high-skilled firms find it unprofitable to organize information production by hiring analysts, so that the economy transitions to the decentralized equilibrium, where all information production and trading are carried out by stand-alone

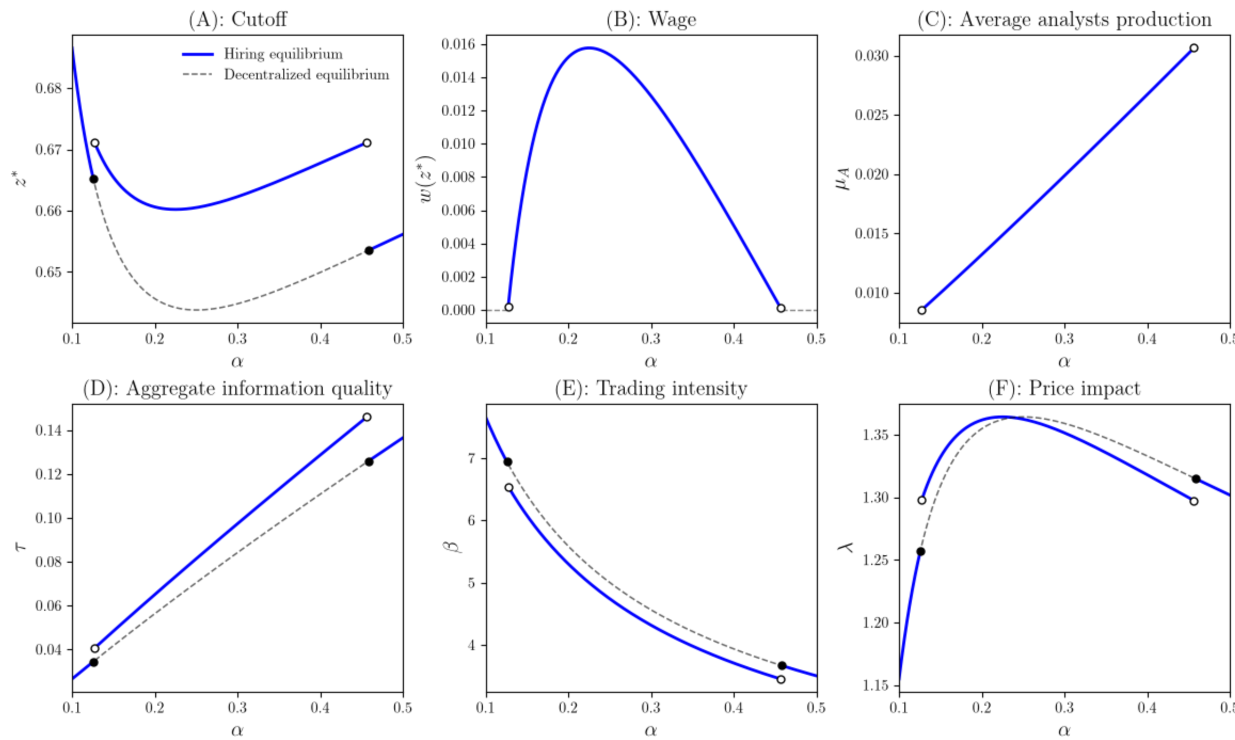

Figure 4: Phase transition

Note: This figure plots the equilibrium variables for the entire range of $\alpha \in [\alpha_0, 1]$. The parameter setting is the same as that used in Figure 1.

managers without forming analyst teams. This result implies that organized information production ceases to exist not only when the technology level is low but also when it is sufficiently high. Moreover, as discussed below, the phase transition involves abrupt changes in the labor market and information production, which in turn feeds back to the financial market variables.

4.2.2 Discontinuous Phase Transition

When the economy moves from the hiring to the decentralized equilibrium, both at $\alpha = \alpha_l$ and $\alpha = \alpha_h$, the cutoff z^* exhibits a discontinuous jump. This result arises from three competing effects of the labor market breakdown.

Firstly, the labor market in the hiring equilibrium serves as the outside option for low-skilled informed agents ($z < z^*$): they do not exert effort, but if they successfully acquire financial knowledge with probability ϕ, they can participate in the labor market as analysts and receive wage $w > 0$. This backup option does not exist in the decentralized equilibrium, increasing the relative value of being a stand-alone manager and encouraging agents to exert effort. However, the impact of this effect is not significant at the thresholds, as the wage level and the value of the outside option are zero, i.e., $w(z^*) = 0$ at $\alpha = \alpha_h$ and $\alpha = \alpha_l$.

Secondly, the absence of the outside option for informed low-skilled agents causes an informational loss: in the hiring equilibrium, they contribute to information production within trading firms, while this information is lost in the decentralized equilibrium. This effect is captured by $\tau(z^*, \alpha) > \tau_d(z_d^*, \alpha)$ in equations (3.17) and (3.20). Hence, in the decentralized equilibrium, trading firms bring less information into the financial market, competition among them is mitigated, and the price impact becomes lower, allowing every surviving firm to trade more aggressively for a given skill level, as $\beta(\tau_d(z^*, \alpha)) > \beta(\tau(z_d^*, \alpha))$ attests. Since individual firms' profit improves as the economy switches from the hiring equilibrium to the decentralized equilibrium, the cutoff z^* declines, and more agents start exerting effort. This effect is captured by the LHS of condition (4.3). As the strict inequality $\tau(z^*, \alpha) > \tau_d(z_d^*, \alpha)$ holds even when $w(z^*) \to 0$, this effect remains significant at the thresholds, contributing to the discontinuity in phase transitions.

Thirdly, firms in the decentralized equilibrium cannot hire analysts to expand information production even when hiring analysts is marginally profitable. Thus, a transition from the hiring to the decentralized equilibrium reduces firms' expected profits and raises the bar for knowledge acquisition, z^*. This channel also contributes to the discontinuity at the phase transition, as the marginal benefit of hiring analysts remains positive ($q^* > 0$) even when $w = 0$, while they do not organize information production because doing so leads to negative utility net of the participation cost, κ. Overall, condition (4.3) reflects the competition between the second negative effect and the third positive effect of the labor market breakdown on the cutoff z^*.

4.2.3 Financial Market

As shown in the lower panels of Figure 4, the financial market variables experience an abrupt change at the phase transition. The decline in the cutoff z^* at the transition from the hiring to the decentralized equilibrium causes a discontinuous decline in the total information quality

(τ), and trading firms start trading more aggressively in a discontinuous manner.

Interestingly, the reaction of the price impact λ (and thus market illiquidity) is asymmetric between the lower ($\alpha = \alpha_l$) and the upper ($\alpha = \alpha_h$) thresholds. At the lower threshold, the aggregate information quality is relatively low, and λ responds positively when the order flow becomes marginally more informative. In contrast, at the upper threshold, the order flow reveals too much information about the asset fundamentals. The market maker seeks to avoid incorporating abundant information and reduces λ when the information quality marginally improves. This asymmetric reaction of the market maker across high- and low-information quality regimes causes λ to jump in opposite directions at the two thresholds, even though the cutoff z^* jumps in the same direction.

4.3 Implications

4.3.1 Technological Progress and Organized of Information Production

Two seemingly incompatible observations describe the organizational consequences of information technology. One of the observations is that AI and other GPTs democratize financial information production by expanding what individual traders can accomplish, allowing smaller teams to compete with large research organizations (e.g., Chang, Dong, Margin, and Zhou, 2026). The other emphasizes increasing concentration, as large financial firms attract talent and exploit technological advances at a scale that smaller firms cannot match (e.g., Rosen, 1981; Autor, Dorn, Katz, Patterson, and Van Reenen, 2020; Business Insider, 2025). Existing theories of information processing within organizations typically discipline organization size through internal costs of communication or coordination (e.g., VanZandt, 1999; Meagher and VanZandt, 1998; Garicano, 2000; Stein, 2002; Acemoglu, Aghion, Lelarge, Van Reenen, and Zilibotti, 2007) and therefore provide limited scope for these opposing forces to operate at different stages of technological development.

My model reconciles these two narratives by placing the disciplining force outside the organization, i.e., the competitive financial market. At relatively low levels of technological development ($\alpha < \alpha^*$), the capability effect dominates: improvements in individual information-processing capacity induce intermediate-skilled agents to establish trading firms rather than supply their capacity to incumbent firms. Information production therefore becomes more decentralized, with more firms operating smaller analyst teams. Beyond a certain technology level ($\alpha \geq \alpha^*$), however, the same technological progress intensifies competition and erodes informational rents, inducing marginal firms to exit and leaving information production concentrated among fewer, more skilled firms with larger analyst teams. Finally, when technology becomes sufficiently advanced, organized information production becomes unprofitable and the hiring equilibrium collapses, generating a second form of decentralization through stand-alone trading (Proposition 12). Thus, technological progress can generate decentralization, concentration, and ultimately decentralization again.

The two forms of decentralization are observationally similar in organizational structure but fundamentally different in their economic implications. Capability-driven decentralization occurs while informational rents are rising, whereas competition-driven decentralization occurs as informational rents are being compressed. Both mechanisms improve aggregate information quality (τ increases) and induce informed trading firms to trade less aggressively (β declines), but they imply opposite responses of price impact: λ rises under capability-driven decentralization and falls under competition-driven decentralization. Hence, organizational structure alone cannot distinguish the two forces; its joint evolution with price impact provides a simple empirical criterion for doing so.

4.3.2 Concentration versus Decentralization

The middle phase of the organizational evolution (i.e., competition-driven concentration) is consistent with the broader literature on superstar firms, which emphasizes that technological progress can disproportionately benefit the most productive organizations (e.g., Rosen, 1981; Autor, Dorn, Katz, Patterson, and Van Reenen, 2020).[21] In my model, intensifying competition similarly eliminates marginal firms and concentrates information production among fewer, more skilled organizations with larger analyst teams.

Existing theories typically imply a monotonic relationship between technological progress and market concentration. The novel implication of my result is that concentration is not necessarily permanent: once technology becomes sufficiently advanced, further improvements

[21] See Aquilina, Budish, and O'Neill (2021) for related dynamics in high-frequency financial markets.

erode informational rents to the point that organized information production can no longer cover its fixed participation cost. The economy therefore transitions discontinuously back to decentralized production of information. In this sense, a decline in concentration need not indicate weaker technology or lower information quality, but it may instead signal that information has become sufficiently abundant that organizing its production is no longer economically viable.

5 Welfare and Policy Implications

Under the hiring equilibrium, an agent's ex-ante expected surplus is defined by

$$U_h = \int_{z^*}^{1} \mathrm{E}[U_{z,M}]dz + \int_0^{z^*} w dz,$$

where the subscript "h" represents the hiring equilibrium. Namely, if she draws a high skill ($z \geq z^*$), she exerts effort and becomes a firm, earning $\mathrm{E}[U_{z,M}]$ in expectation. If, in contrast, her skill is low ($z < z^*$), she does not exert effort and becomes an analyst, receiving wage w. Applying equation (3.12), the definition of the cutoff z^* in (3.13), and the labor market clearing condition ($Q = S = z^*$), the welfare function is simplified as follows:

$$\begin{aligned} U_h &= \int_{z^*}^{1} \left(\alpha\rho\sigma_v^2\beta^2(\tau(z^*,\alpha))z + \frac{\left(\alpha\rho\sigma_v^2\beta^2(\tau(z^*,\alpha))\frac{\phi z^*}{2} - w\right)^2}{2\gamma} - \kappa \right) dz + \int_0^{z^*} w dz \\ &= \alpha\rho\sigma_v^2\beta^2(\tau(z^*,\alpha))\frac{(1-z^*)^2}{2} + w. \end{aligned} \tag{5.1}$$

In this expression, w in the second term represents the reservation utility that an agent always receives regardless of her occupation, and she obtains additional utility in the first term if she becomes a firm.

In contrast, if the economy is in the decentralized equilibrium (indicated by the subscript

“d”), the expected surplus is expressed as

$$\begin{aligned} U_d &= \int_{z_d^*}^{1} \left(\alpha\rho\sigma_v^2\beta^2(\tau_d(z_d^*,\alpha))z - \kappa\right) dz \\ &= \alpha\rho\sigma_v^2\beta^2(\tau_d(z_d^*,\alpha))\frac{(1-z_d^*)^2}{2}. \end{aligned} \tag{5.2}$$

Intuition follows from (5.1), while the profit margin, $R_d(z_d^*,\alpha) \equiv \alpha\rho\sigma_v^2\beta^2(\tau_d(z_d^*,\alpha))$, is defined using variables in Proposition 6, and the wage income is not available due to the absence of labor-market activity, and the cutoff is characterized by z_d^*.

Finally, the noise trader obtains the following expected utility:

$$\begin{aligned} U_n &= E[(v-p)u] \\ &= -\lambda\sigma_u^2, \end{aligned} \tag{5.3}$$

as she executes her order of size u and incurs the price impact λ in doing so. The functional form of U_n is common across the two equilibria, while the price impact λ differs across the equilibria. Because the financial market is a zero-sum game, aggregate welfare including the noise trader is less informative in this setting; the following analyses therefore focus on the welfare of agents who organize information production and participate in the financial market.

5.1 Welfare Implications of Organized Information Production

A natural question is whether the endogenous emergence of organized information production improves agents' welfare. The following result compares agents' welfare in the hiring equilibrium with that in the hypothetical decentralized equilibrium at the same technology level α, showing that organized production triggers two competing welfare effects, and that their interaction leaves the overall impact ambiguous.

Proposition 13. *When the hiring and decentralized equilibria coexist,*

$$U_d - U_h = \underbrace{\frac{1}{2}R_d(z_d^* - z^*)^2 + \frac{1}{2}(1-z^{*2})(R_d - R)}_{\Delta_{u,1}} - \underbrace{\frac{z^{*2}}{2}\left(\phi R - \frac{\gamma}{1-z^*}\right)}_{\Delta_{u,2}}, \tag{5.4}$$

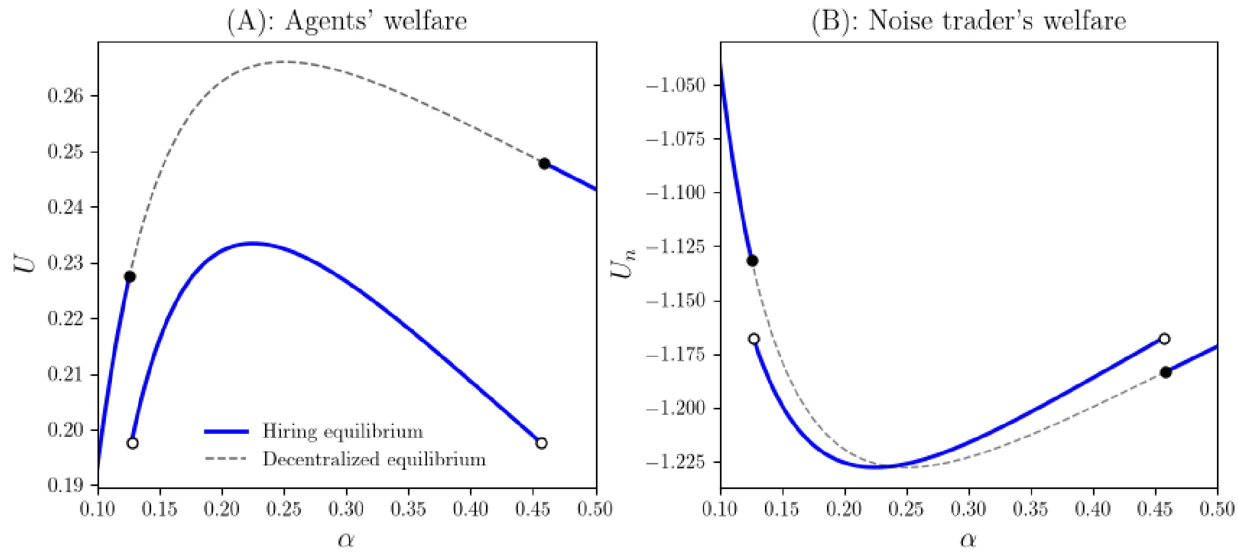

Figure 5: Welfare

Note: This figure plots an agent's expected welfare and that of the noise trader. The parameter setting is the same as that used in Figure 1.

where $R = R(z^*, \alpha)$ *and* $R_d = R_d(z_d^*, \alpha)$ *represent the profit margins in the two equilibria, respectively, and both* $\Delta_{u,1}$ *and* $\Delta_{u,2}$ *in (5.4) are strictly positive.*

Terms summarized by $\Delta_{u,1}$ in (5.4) capture a competitive externality: pooling analysts' information raises the total information reaching the financial market, and since no individual firm accounts for information externality, everyone's trading margin in the hiring equilibrium is compressed relative to the decentralized equilibrium. In contrast, $\Delta_{u,2}$ captures the direct trading-profit gain from putting analysts' information to use, net of the cost of running the organization. This is exactly the private surplus that encourages organized information production, and it is strictly positive whenever hiring is viable. Whether organized information production raises or lowers welfare comes down to whether this private surplus $\Delta_{u,2}$ outweighs the externality it creates through the financial market competition $\Delta_{u,1}$.

Neither effect dominates in general, suggesting that the endogenous emergence of organized information production can reduce the welfare of the very agents who choose to organize it. Figure 5 plots such a situation, demonstrating $U_d > U_h$ in $\alpha \in (\alpha_l, \alpha_h)$. Agents would be collectively better off avoiding the labor market for analysts altogether, even though forming it is individually optimal for both the marginal manager and the marginal analyst, as they do not internalize the information externality. Organized information production, in other words, can be inefficient from the perspective of agents' total welfare precisely when it is individually rational.

Notably, the analyst wage itself plays no role in the above comparison: it is paid by managers and received by analysts, so the wage transfer cancels out in aggregate agent welfare. The intuition that hiring "shares the gains" with analysts through wages is therefore misleading: what actually determines the comparison is the competitive externality and the private surplus from organizing information production.

This result contrasts with the traditional view that specialization and the division of labor improve welfare by increasing productive efficiency in general production environments (e.g., Becker and Murphy, 1992). Within the context of information production, a related view holds that organizing information processing into hierarchies raises productivity (e.g., VanZandt, 1999; Garicano, 2000). My model suggests that this need not hold once information is produced for trading in a competitive financial market, as organizing information production intensifies competition and can erode informational rents more than it creates the private surplus. This result complements Stein (2002), where whether centralized or decentralized information production is more efficient depends on the type of information and its communication cost within the firm, while the inefficiency in my model instead arises from a different, market-based mechanism.

5.2 Policy Implications

Broadly speaking, governments can influence the organization of financial information production through two channels. The first is technology policy that shifts the economy's information-processing frontier directly, including subsidies, infrastructure support, or restrictions on AI use in trading and investment advice, such as those raised in the SEC's 2023 proposal on predictive data analytics (SEC, 2023). The second is organizational policy, which instead targets firms' incentives to organize information production through, for example, employment regulation or rules governing how research is bundled with trading services.

My model suggests technology policy is unlikely to produce monotonic welfare gains: because agents' and the noise trader's welfare responds non-monotonically to technological progress, subsidizing the adoption of AI and other technologies redistributes welfare between agents and the noise trader in a way that itself reverses as technology matures, rather than delivering uniform gains to either side.

Organizational policy operates differently. For example, the MiFID II research-unbundling rules required asset managers to pay for investment research separately from trading commissions, raising the effective cost of maintaining research relationships. The empirical literature reports a decline in sell-side analyst coverage associated with such rules (e.g., Fang, Hope, Huang, and Moldovan, 2020; Guo and Mota, 2021). My model suggests why such a policy need not be purely costly: by weakening firms' incentive to organize information production, it can push the economy from the hiring equilibrium toward the decentralized equilibrium, a shift that can benefit agents in the financial market when organizing information production would otherwise be an equilibrium outcome.

6 Conclusion

This paper studies how technological progress reshapes the organization of financial information production through the interaction between labor markets and financial markets. I show that technological progress can generate multiple and opposing organizational changes: it initially decentralizes information production by expanding individual capabilities, then induces concentration by intensifying competition among informed firms, and eventually leads to a second form of decentralization when organized information production becomes unprofitable. The key mechanism is that competitive information externalities affect not only the profitability of information production but also the organizational structure through which information is produced. These results highlight that the relationship between technology and organizational structure is more complex than the conventional view that technological progress simply empowers individuals.

Several extensions remain for future research. The model abstracts from a number of features of real-world financial markets, including heterogeneous technologies across firms and strategic technology adoption. Future work could examine how firms endogenously choose whether and how to adopt new information technologies, and how such decisions interact with market competition and organizational design. More broadly, understanding how general-purpose technologies reshape the boundary between individuals and organizations remains an important question for the future of information-intensive industries.

Appendix

A Proof of Propositions 5 and 6

If the hiring equilibrium exists, the cutoff z^* is the solution to $G(z,\alpha)=0$ with

$$G(z,\alpha) = R(z,\alpha)z\left(1-\frac{\phi}{2}\right) + \gamma\frac{z\left(1-\frac{z}{2}\right)}{(1-z)^2} - \kappa, \tag{A.1}$$

where $R(z,\alpha)=\alpha\rho\sigma_v^2\beta(\tau(z,\alpha))$, and β is the unique solution to $B(\beta,\tau(z))=0$ with

$$B(\beta,\tau(z)) = \rho\beta(\beta^2\tau^2(z)\sigma_v^2+\sigma_u^2)-\sigma_u^2. \tag{A.2}$$

Since $\frac{\partial B}{\partial \beta}>0$ and $\frac{\partial B}{\partial z} = -\alpha(1-\phi)z\frac{\partial B}{\partial \tau}<0$, the implicit function theorem implies $\frac{d\beta}{dz}>0$. Therefore, $\frac{\partial G}{\partial z}>0$, $G(0,\alpha)=-\kappa<0$, and $G(1,\alpha)=\infty$ hold, suggesting that $G=0$ has a unique solution $z^*\in(0,1)$. Given z^*, the labor market clearing condition yields the following wage:

$$w(z^*) = \frac{\phi}{2}R(z^*,\alpha)z^* - \gamma\frac{z^*}{1-z^*}.$$

If $w(z^*)>0$, then z^* is the hiring equilibrium cutoff. Otherwise, there is no hiring equilibrium.

The decentralized equilibrium is defined by the cutoff z_d^* as the solution to $G_d(z,\alpha)=0$, where

$$G_d(z,\alpha) = R(z,\alpha)z-\kappa. \tag{A.3}$$

The definition of R is the same as above, while $\beta(\tau_d(z))$ involves $B(\beta,\tau_d(z))=0$ with $\tau_d(z)=\frac{\alpha(1-z^2)}{2}$. Due to the same arguments as the hiring equilibrium, $\frac{\partial G}{\partial z}>0$. It also holds that $G_d(0,\alpha)=-\kappa<0$ and $G_d(1,\alpha)=R(1,\alpha)$. Note that $\tau_d(1)=0$ and $\beta(\tau_d(1))=\frac{1}{\rho}$, leading to $G_d(1,\alpha)=\frac{\alpha\sigma_v^2}{\rho}-\kappa$. The solution to $G_d(z,\alpha)=0$ exists in $z\in(0,1)$ if, and only if, $G_d(1,\alpha)>0$, that is, $\alpha>\frac{\rho\kappa}{\sigma_v^2}$. Since Assumption 2 ensures that $\alpha_0>\frac{\rho\kappa}{\sigma_v^2}$, the decentralized equilibrium always exists.

B Proof of Lemma 8 and Propositions 9–11

As $\frac{\partial R}{\partial z} > 0$ holds, it suffices to evaluate the following partial derivative:

$$\frac{\partial R}{\partial \alpha} = \rho \sigma_v^2 \left(\beta(\tau(z^*, \alpha)) + 2\alpha \frac{\partial \beta(\tau(z^*, \alpha))}{\partial \alpha} \right). \tag{B.1}$$

Note that $\frac{\partial \beta(\tau(z^*,\alpha))}{\partial \alpha} = \frac{\partial \tau}{\partial \alpha}\frac{\partial \beta(\tau)}{\partial \tau}$, where equation (3.17) implies that

$$\frac{\partial \tau}{\partial \alpha} = \frac{1}{2}(1 - (1-\phi)z^{*2}) = \frac{\tau}{\alpha}. \tag{B.2}$$

Also, from equation (A.2), $\tau^2 = \frac{\sigma_u^2}{\sigma_v^2}\frac{1-\rho\beta}{\rho\beta^3}$. By taking the partial derivative of both sides with respect to α,

$$2\tau \frac{\partial \tau}{\partial \alpha} = -\frac{\sigma_u^2}{\sigma_v^2}\frac{(3-2\rho\beta)}{\rho\beta^4}\frac{\partial \beta}{\partial \alpha}.$$

Applying (B.2) and rearranging for $\partial\beta/\partial\alpha$ yields

$$\frac{\partial \beta}{\partial \alpha} = -\frac{2\beta(1-\rho\beta)}{\alpha(3-2\rho\beta)} < 0.$$

Then, the partial derivative (B.1) becomes

$$\frac{\partial R}{\partial \alpha} = \frac{\rho\sigma_v^2 \beta(\tau(z^*,\alpha))}{3 - 2\rho\beta(\tau(z^*,\alpha))} \left(2\rho\beta(\tau(z^*,\alpha)) - 1 \right).$$

Since $\beta < 1/\rho$ holds in the equilibrium, the numerator is positive, leading to

$$\frac{\partial R}{\partial \alpha} \gtrless 0 \Leftrightarrow \beta(\tau(z^*,\alpha)) \gtrless \frac{1}{2\rho}.$$

As β is the solution to $B = 0$ with $\frac{\partial B}{\partial \beta} > 0$, the last inequality is re-written as follows:

$$\begin{aligned} \beta > \frac{1}{2\rho} \Leftrightarrow B(1/2\rho, \alpha) &= \frac{1}{2}\left(\frac{1}{4\rho^2}\tau(z^*,\alpha)^2\sigma_v^2 + \sigma_u^2 \right) - \sigma_u^2 < 0, \\ &\Leftrightarrow \tau(z^*,\alpha) < 2\rho\frac{\sigma_u}{\sigma_v} \\ &\Leftrightarrow z^* > z_0(\alpha), \end{aligned} \tag{B.3}$$

where

$$z_0(\alpha) \equiv \begin{cases} z_0^+(\alpha) = \sqrt{\frac{1-\frac{1}{\alpha}\frac{4\rho\sigma_u}{\sigma_v}}{1-\phi}} & \text{if } \alpha \geq \frac{4\rho\sigma_u}{\sigma_v}, \\ 0 & \text{if } \alpha < \frac{4\rho\sigma_u}{\sigma_v}. \end{cases}$$

By the implicit function theorem, z^* is decreasing in α if $z^* > z_0(\alpha)$, and vice versa.

Firstly, if $\alpha > \frac{4\rho\sigma_u}{\phi\sigma_v}$, then $\alpha > \frac{4\rho\sigma_u}{\sigma_v}$ holds, and $z_0^+ > 1$ implies that condition (B.3) is always violated. In this case, $\beta < 1/2\rho$ always holds, and thus z^* is increasing in α. Secondly, if $\alpha \leq \frac{4\rho\sigma_u}{\sigma_v}$, then $z_0(\alpha) = 0$, suggesting that $z^* > z_0(\alpha)$ always holds. Thus, in this case, z^* is decreasing in α. Thirdly, consider the case with $\frac{4\rho\sigma_u}{\sigma_v} < \alpha \leq \frac{4\rho\sigma_u}{\phi\sigma_v}$. In this case, $z_0(\alpha) = z_0^+(\alpha) < 1$. Note that z_0^+ is monotonically increasing in α, while z^* is increasing in α if and only if $z^* < z_0$. Thus, consider $\alpha = \alpha^*$ such that $z^*(\alpha^*) = z_0^+(\alpha^*)$. If such α^* exists, then it is a uniquely determined, as $\alpha < \alpha^*$ implies $z^* > z_0$ and vice versa, i.e., z^* and z_0 crosse at most once. As$z^*(\alpha^*) = z_0^+(\alpha^*)$ is equivalent to $2\rho\beta = 1$, equation (A.1) implies that α^* solves $g(\alpha^*) = 0$ with

$$g(\alpha) \equiv \alpha\sigma_v^2 z_0^+(\alpha)\left(1 - \frac{\phi}{2}\right) + 4\rho\gamma\frac{z_0^+(\alpha)\left(1 - \frac{z_0^+(\alpha)}{2}\right)}{\left(1 - z_0^+(\alpha)\right)^2} - 4\rho\kappa. \tag{B.4}$$

g is monotonically increasing in α^*. Also, as $\alpha^* \to \frac{4\rho\sigma_u}{\phi\sigma_v}$, it holds that $z_0^+ \to 1$, and thus $g \to \infty$; as $\alpha^* \to \frac{4\rho\sigma_u}{\sigma_v}$, it holds that $z_0^+ \to 0$, and thus $g \to -4\rho\kappa < 0$. Therefore, within the interval $\frac{4\rho\sigma_u}{\sigma_v} < \alpha \leq \frac{4\rho\sigma_u}{\phi\sigma_v}$, $g(\alpha) = 0$ always has a unique solution, which I define α^*. Following the arguments so far,

$$\frac{dz^*}{d\alpha} \lesseqgtr 0 \Leftrightarrow \alpha \lesseqgtr \alpha^*.$$

By the implicit function theorem, $\frac{\partial R}{\partial \alpha} \gtreqless 0 \Leftrightarrow \frac{dz^*}{d\alpha} \lesseqgtr 0$. This concludes the proof of Lemma 8 and Proposition 9. As $\mu_A = \frac{\phi z^*}{2}$, the above inequality implies point (ii) of Proposition 10.

From the equilibrium condition (A.1),

$$\begin{aligned} R(z^*, \alpha) &= \alpha\rho\sigma_v^2\beta^2(\tau(z^*, \alpha))z^* \\ &= \frac{2\kappa - \gamma\left(\frac{1}{(1-z^*)^2} - 1\right)}{2 - \phi} \\ &\equiv \bar{R}(z^*), \end{aligned}$$

where α does not directly appear in the last expression. From equation (3.13), wage $w(z^*)$ satisfies

$$\bar{R}(z^*) + \frac{\left(\bar{R}(z^*)\frac{\phi_l}{2} - w\right)^2}{2\gamma} = w + \kappa,$$

so that α affects w only through changes in z^*. It holds that $\frac{dw}{d\alpha} = \frac{dz^*}{d\alpha}\frac{dw}{dz^*}$, and the implicit function theorem implies

$$\frac{\partial w}{\partial z^*} = \frac{d\bar{R}}{dz^*}\frac{\gamma + \left(\bar{R}(z^*)\frac{\phi_l}{2} - w\right)\frac{\phi_l}{2}}{\bar{R}(z^*)\frac{\phi_l}{2} - w + \gamma} < 0,$$

where the last inequality comes from $\frac{d\bar{R}}{dz^*} < 0$. Therefore, $\frac{dw}{d\alpha} \gtreqless 0 \Leftrightarrow \frac{dz^*}{d\alpha} \lesseqgtr 0$, and w exhibits the opposite reaction to changes in α compared to z^*, attesting point (i) of Proposition 10.

As for Proposition 11, equation (3.17) implies that

$$\frac{d\tau}{d\alpha} = \frac{\partial\tau}{\partial\alpha} - \alpha(1-\phi)z^*\frac{dz^*}{d\alpha}.$$

If $\frac{dz^*}{d\alpha} < 0$, then $\frac{d\tau}{d\alpha} > 0$ holds. Suppose that $\frac{dz^*}{d\alpha} > 0$. Note that $z^*(\alpha)$ satisfies $G(z^*(\alpha), \alpha) = 0$. By taking the total derivative of this condition with respect to α and rearranging for $\frac{d\tau}{d\alpha}$, I obtain

$$\begin{aligned}\alpha\rho\sigma_v^2 z^*(\alpha)\left(1 - \frac{\phi}{2}\right)\frac{d\tau}{d\alpha}\frac{d\beta}{d\tau} =& -\frac{R(z^*(\alpha),\alpha)}{\alpha}z^*(\alpha)\left(1-\frac{\phi}{2}\right) \\ & - \frac{dz^*}{d\alpha}\frac{\partial}{\partial z^*}\left[R(z^*(\alpha),\alpha)z^*(\alpha)\left(1-\frac{\phi}{2}\right) + \gamma\frac{z^*(\alpha)\left(1-\frac{z^*(\alpha)}{2}\right)}{\left(1-z^*(\alpha)\right)^2}\right] \\ & < 0.\end{aligned}$$

where the last inequality comes from the assumption that $\frac{dz^*}{d\alpha} > 0$. Since $\frac{d\beta}{d\tau} < 0$, it holds that $\frac{d\tau}{d\alpha} > 0$. Moreover, from equation (A.2), $\tau^2 = \frac{\sigma_u^2}{\sigma_v^2}\frac{1-\rho\beta}{\rho\beta^3}$ holds, suggesting that $\frac{d\tau}{d\alpha} \propto \frac{d\beta}{d\alpha}$. Thus, $\frac{d\beta}{d\alpha} > 0$. The behavior of λ is computed from (3.5) as

$$\frac{d\lambda}{d\alpha} = \frac{d(\beta\tau)}{d\alpha}\frac{\sigma_u^2 - \beta^2\tau^2\sigma_v^2}{(\beta^2\tau^2\sigma_v^2 + \sigma_u^2)^2}\sigma_v^2.$$

Since $\frac{d(\beta\tau)}{d\alpha} > 0$, it holds that $\frac{d\lambda}{d\alpha} > 0 \Leftrightarrow \beta\tau < \frac{\sigma_u}{\sigma_v}$. From (A.2), $\beta\tau = \frac{\sigma_u}{\sigma_v} \Leftrightarrow 2\rho\beta = 1$.

Together with the fact that $\beta\tau$ is monotonically increasing in α, this identity implies that λ is hump-shaped with respect to α with the tipping point being α^*.

C Proof of Proposition 12

C.1 The Existence of the Hiring Equilibrium

The hiring equilibrium exists when the solution z^* to $G(z^*, \alpha) = 0$ supports $w > 0$, where Proposition 10 attests that w takes a hump-shaped curve against α with its peak being $w^* \equiv w(z^*(\alpha^*))$. $G = 0$ implies that

$$\begin{aligned} w(\alpha) &\equiv \frac{\phi}{2-\phi}\left(\kappa - \gamma\frac{z^*(\alpha)\left(1-\frac{z^*(\alpha)}{2}\right)}{\left(1-z^*(\alpha)\right)^2}\right) - \gamma\frac{z^*(\alpha)}{1-z^*(\alpha)} \\ &= \frac{Y(z^*(\alpha))}{(2-\phi)(1-z^*(\alpha))^2}, \end{aligned}$$

where

$$Y(z^*(\alpha)) = \left(\phi\kappa + 2\gamma(1-\frac{\phi}{4})\right) z^{*2}(\alpha) - 2(\phi\kappa+\gamma)z^*(\alpha) + \phi\kappa.$$

At $\alpha = \alpha^*$, it holds that $2\rho\beta = 1$, leading to $z^*(\alpha^*) = z_0^+(\alpha^*) = \sqrt{\frac{1-\frac{1}{\alpha^*}\frac{4\rho\sigma_u}{\sigma_v}}{1-\phi}}$, as defined in Appendix B. Hence, the maximum wage is given by

$$w^* = \left(\phi\kappa + 2\gamma(1-\frac{\phi}{4})\right) z_0^{+2}(\alpha^*) - 2(\phi\kappa+\gamma)z_0^+(\alpha^*) + \phi\kappa.$$

If $w^* > 0$, then the continuity of w^* regarding α implies that there is an interval (α_l, α_h) such that $w(z^*(\alpha)) > 0$.

Consider a quadratic function

$$T(z) = \left(\frac{\phi}{2}\kappa + \gamma(1-\frac{\phi}{4})\right) z^2 - (\phi\kappa+\gamma)z + \frac{\phi}{2}\kappa,$$

so that $w^* = 2T(z_0^+(\alpha^*))$. The discriminant of T is $D = (\phi\kappa+\gamma)^2 - 4\frac{\phi}{2}\kappa\left(\frac{\phi}{2}\kappa + \gamma\left(1-\frac{\phi}{4}\right)\right) = \frac{\gamma\phi^2\kappa}{2} + \gamma^2 > 0$. Also, $T(0) > 0$ and $T(1) = -\gamma\frac{\phi}{4} < 0$, suggesting that $T = 0$ has a unique solution in $z \in (0,1)$, which is

$$z_{w=0} = \frac{\phi\kappa + \gamma - \sqrt{\frac{\gamma\phi^2\kappa}{2} + \gamma^2}}{\phi\kappa + 2\gamma\left(1 - \frac{\phi}{4}\right)}. \tag{C.1}$$

Therefore, $w^* > 0$ holds if and only if $z_0^+(\alpha^*) < z_{w=0}$, which is equivalent to

$$\alpha^* < \alpha_{w=0} \equiv \frac{1}{1-(1-\phi)z_{w=0}^2}\frac{4\rho\sigma_u}{\sigma_v}.$$

Since α^* is a unique solution to $g = 0$ in equation (B.4), $\alpha^* < \alpha_{w=0} \Leftrightarrow g(\alpha_{w=0}) > 0$, that is,

$$\alpha_{w=0}\sigma_v^2\left(1 - \frac{\phi}{2}\right) z_{w=0} + 4\rho\gamma\frac{z_{w=0}\left(1 - \frac{z_{w=0}}{2}\right)}{\left(1 - z_{w=0}\right)^2} > 4\rho\kappa. \tag{C.2}$$

Moreover, under condition (C.2), the threshold α_j that brings about $w = 0$ are implicitly defined by $Y(z^*(\alpha)) = 0$. By applying $w(z^*(\alpha)) = 0$, this condition is re-written as

$$\alpha_j\beta^2(z_{w=0}, \alpha_j)(1 - z_{w=0}) = \frac{2\gamma}{\phi\rho\sigma_v^2}.$$

C.2 Characterizing the Phase Transition

The decentralized equilibrium solves $G_d(z, \alpha) = 0$ in (A.3), where G_d is monotonically increasing in z. Hence, if

$$G_d(z^*, \alpha) = \alpha\rho\sigma_v^2\beta^2(\tau_d(z^*(\alpha)))z^*(\alpha) - \kappa > 0,$$

then $z^* > z_d^*$ holds. At the threshold α_j, $w = 0$ holds. Thus, the above condition becomes

$$G_{w=0,j} = \alpha_j\rho\sigma_v^2\beta^2(\tau_d(z_{w=0}, \alpha_j))z_{w=0} - \kappa > 0.$$

By eliminating α using $G(z, \alpha) = 0$, the above condition becomes equivalent to condition (4.3).

The following figures illustrate the behavior of equilibrium variables when the phase transition from the hiring to the decentralized equilibrium involves an upward jump in the

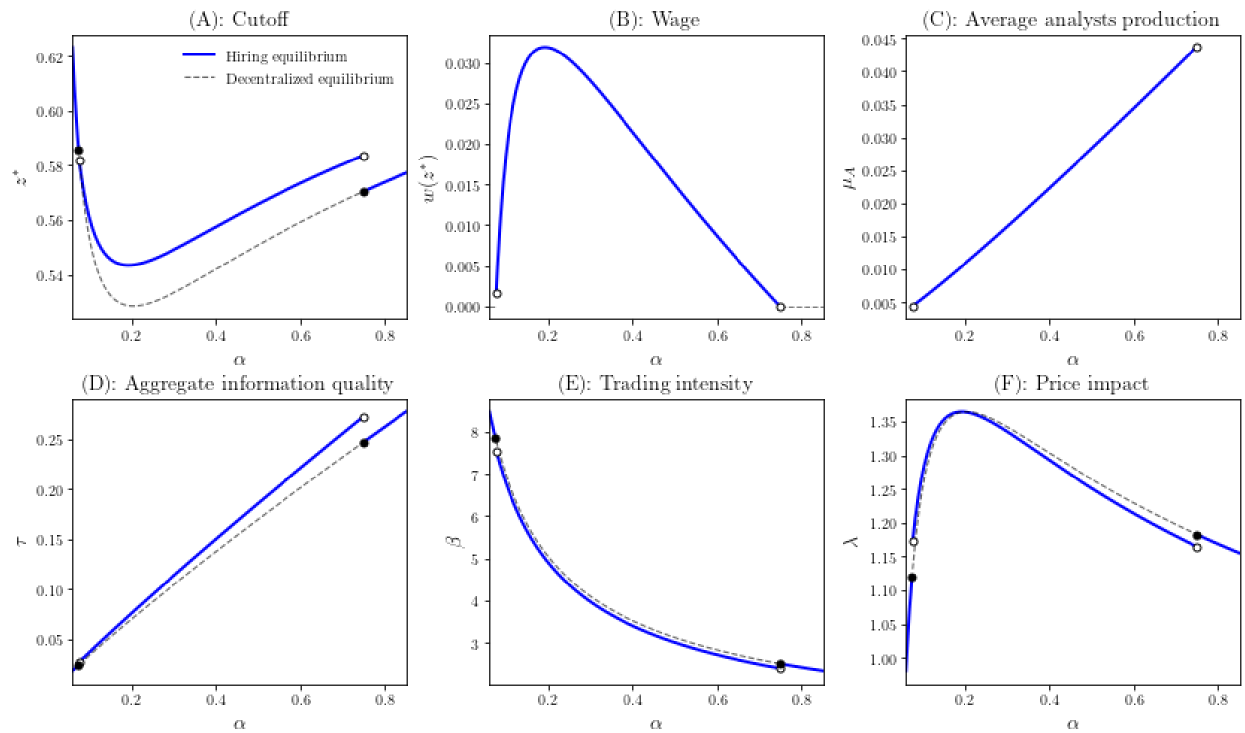

Figure 6: Phase transition

Note: This figure plots the equilibrium variables for the entire range of $\alpha \in [\alpha_0, 1]$. The parameter setting is as follows: $\phi = 0.2, \kappa = 1.8, \gamma = 0.12, \sigma_v^2 = 6.7, \sigma_u^2 = 0.9, \rho = 0.1$.

cutoff z^*.

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